Nabz: Offline Clinical Voice Scribe for Pakistani OPDs

Nov 3, 2026 · @Abdullah Haroon

Summary

Nabz is an offline voice scribe for doctors in Pakistani outpatient clinics. It turns a spoken consultation into the paperwork the visit needs, so the doctor spends those minutes on the patient instead of the pen.

The project pairs a working product, used by real doctors, with a research paper built on that same system. Everything runs offline on open, quantized models, on the doctor's own Android phone or the old PC already on the OPD desk. Patient audio never leaves the room, and neither internet nor a GPU is needed.

The problem

A government OPD doctor sees around 100 patients a day, four times the international standard of 25 per consultant (APP). A 2008 study in Civil Hospital Karachi's surgical OPD measured an average consultation of about 6 minutes (PJMS). The clinical note is the first thing dropped.

Four things make this hard to automate here:

- **Code-switching.** One consultation mixes Punjabi symptoms, English drug brands and Urdu instructions, often inside one sentence. On the UrduSpeech benchmark, Whisper-large-v3 goes from 28.9% WER on monolingual Urdu to 53.2% on code-switched speech (UrduSpeech).
- **Drug names.** DRAP counted about 72,000 registered medicines in 2015, and two different drugs were once sold under the same brand name, Valtec (Dawn). One misheard brand is a wrong prescription.
- **Negation and doses.** Urdu puts the negative at the end of the clause, so a dropped last word turns bukhar nahi hai (no fever) into fever. Doses arrive as phrases like aadhi goli din mein do baar.
- **Privacy and connectivity.** Patient audio sent to a foreign cloud is a consent problem, and many clinics face power cuts, patchy internet and ageing computers.

Handwriting compounds the problem. In Peshawar, 58.5% of 1,097 prescriptions had poor legibility and 89% lacked the doctor's name (PJMS 2014). In a Lahore public hospital that wrote prescriptions by hand, 44% of OPD prescriptions contained a prescribing error and

19.2% carried an overdose (PLoS ONE 2014). A printed, checked prescription answers both.

Doctors here want this help but get no support to adopt it. A 2026 survey of 36 physicians and 43 patients in Lahore (Haroon, 2026) found 94% of physicians expect AI to free time for patient contact, yet 92% insist the doctor keep final control. Among quality-checked responses, 87% said their hospital offers no AI training and 78% reported no technical AI staff.

On the patient side, 42% had stopped or changed treatment without asking their doctor, and none without formal education used AI tools. A useful system must leave the doctor in charge, need no training or IT support, and give patients instructions they can follow without an app.

Commercial scribes now exist: MedIQ claims Urdu, Pashto and Punjabi support with a 60% cut in documentation time (TechJuice), and Boston Health AI launched Hami in 2025 (CW Pakistan). Their accuracy claims are their own. No public benchmark of code-switched Urdu-Punjabi clinical speech exists to check them against, as far as a literature search shows. Nabz fills that gap.

The product

The app listens to the whole consultation, or to the doctor's dictation after it, and shows the transcript live as each word is spoken. A draft is ready before the next patient sits down. Nothing is saved or printed until the doctor approves it.

What the doctor gets

- A structured English note: complaint, history, findings, diagnosis, plan. Every line links to the second of audio it came from, so checking a line is one tap.
- A prescription where each drug is matched to a DRAP-registered product and each dose is range-checked. Doubtful items are flagged, never guessed.
- Prompts for gaps, such as allergies never asked.

What the patient gets

- Plain Urdu instructions: which medicine, how much, when, for how long, and which warning signs mean come back. Printed on a small thermal printer, with the Urdu rendered as an image because these printers carry no Nastaliq font, and shared as an Urdu voice note whenever the phone has signal.

Three recording modes, one pipeline

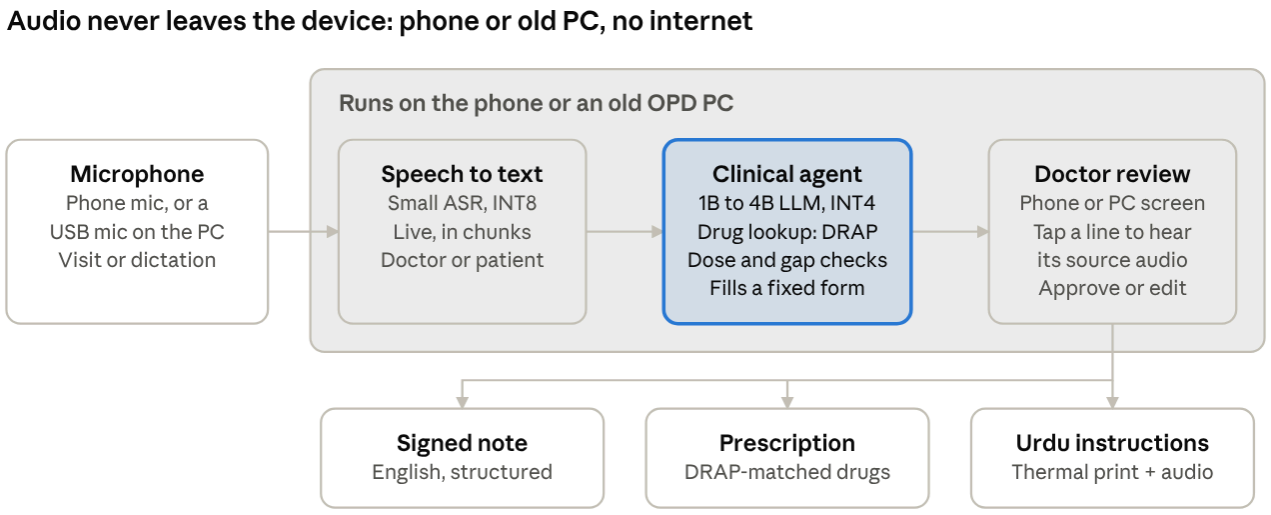
1. Ambient: records the full doctor-patient conversation through the phone on the desk or a USB mic, after the patient's consent is captured on audio. Each line of the live transcript is labelled doctor or patient.
2. Dictation: the doctor speaks a 20 to 60 second summary after the patient leaves, like a WhatsApp voice note. No patient audio is recorded, and it works on any phone.
3. Ambient plus summary: the conversation is recorded, then the doctor adds a 15 second closing summary. The agent treats his summary as the authority and fills in details from the conversation.

The doctor picks the mode per patient. Ambient reuses the dictation pipeline and adds three pieces: chunking at pauses, doctor-voice labelling, and a pass that drops small talk before the note is written.

Where it runs

- Doctor's Android phone: records, transcribes and drafts on the phone itself, with no internet.
- Old OPD PC: the same pipeline as an offline desktop app, which also drives the printer.
- Either works alone. When both are present, the phone hands drafts to the PC over local Wi-Fi for printing. Neither needs a GPU.

System design



system design · microphone, device, outputs

The whole pipeline runs on one device: the doctor's phone or an old OPD PC. Nothing reaches a patient until the doctor approves it.

Stage	On the phone	On an old PC
Speech to text	Whisper base or small at INT8 via whisper.cpp or sherpa-onnx; Omnilingual 300M CTC if its ONNX export holds up. Live transcript from 5 to 8 s chunks cut at pauses by Silero VAD	Whisper small or large-v3-turbo at INT8 via faster-whisper or whisper.cpp, same chunked live transcript
Doctor or patient	Doctor enrolls his voice once; each ambient chunk is matched to it with sherpa-onnx speaker embeddings	Same voice match; full sherpa-onnx diarization when a third voice joins
Note writer	1B to 2B LLM at INT4 (Gemma 4 E2B or a Qwen3.5 small model) via llama.cpp	Up to 4B at INT4 via llama.cpp
Drug, dose and gap checks	Plain code over a local SQLite copy of the DRAP registry	Same
Urdu voice out	MMS-TTS Urdu via sherpa-onnx, or Matcha Urdu TTS in ONNX	Same
App	Android app in Kotlin or Flutter	Offline desktop app with a browser UI, one installer

Training needs a GPU; deployment does not. LoRA and QLoRA fine-tuning up to 4B fits on an 8 GB consumer GPU, and 7B to 8B runs go to a 16 GB RTX A4000. Every component is open source and free.

Running on phones and old PCs

Nabz is built for the hardware doctors already have: an Android phone and the ageing, GPU-less PC on the OPD desk. Android held 91.57% of mobile OS share in Pakistan in October 2026 (StatCounter), so the phone build comes first.

Tier	Typical hardware	Budget for Nabz	Target
Doctor's phone	Android, 6 to 8 GB RAM	Under 2.5 GB RAM, about 2 GB storage	Live transcript in both modes; draft within 60 s of a 30 s dictation, 90 s after a 5 min conversation

Tier	Typical hardware	Budget for Nabz	Target
Low-end phone	Android, 4 GB RAM	Under 1.5 GB RAM	Live dictation; ambient transcribed after the visit, smallest models
Old OPD PC	Core i3 or i5, 4th to 8th gen, 8 GB RAM, no GPU	Under 4 GB RAM	Live transcript and draft within 60 s in both modes

These budgets are design targets. The benchmark measures whether each tier meets them.

What makes it fit

1. **Short audio.** Dictation clips run 20 to 60 seconds and conversations 4 to 5 minutes. Community benchmarks on a Raspberry Pi 5, a rough lower bound for an old office PC, show whisper.cpp running Whisper tiny faster than real time, base at about real time, and small slower (PromptQuorum). Short clips keep even the slower models usable. For conversations, a small model drives the live view and a larger one reruns in the background for the final note.
2. **Stream while recording.** Transcribe 5 to 10 second chunks as the doctor speaks, so only the last chunk is left when recording stops.
3. **One model in memory at a time.** Load speech recognition, unload it, then load the note writer. Peak RAM is the larger model, not the sum.
4. **Small model, strong tools.** Drug lookup, dose ranges and gap checks are plain code over a local copy of the DRAP registry. The LLM only fills a fixed note template, a job a 1B to 2B model can do. It never has to remember drug facts.
5. **Constrained output.** Grammar-constrained decoding in llama.cpp forces every note into valid JSON, so a small model cannot return a broken note.
6. **Single-pass recognition for weak CPUs.** Omnilingual ASR includes a 300M model built for low-power devices (The Decoder), and its CTC variants emit the transcript in one pass rather than token by token.
7. **Builds for old CPUs.** Ship fallback builds for processors without AVX2, which covers PCs built before 2013 and many budget Pentium and Celeron chips.
8. **Integer-only quantization on phone NPUs.** MobileQuant cut on-device LLM latency and energy by 20 to 50% against earlier on-device methods (arXiv).
9. **Power cuts.** The phone keeps working on battery when the PC goes dark.
10. **Offline install.** One APK and one PC installer, no Docker. Model files load once by USB or local Wi-Fi.

One runtime covers most of the speech stack: sherpa-onnx runs offline speech recognition including Whisper, Matcha and VITS text-to-speech, and speaker diarization on Android and desktop (pub.dev). VOSK, a common offline choice, ships no Urdu model (VOSK), so it is left out.

Research questions and contributions

The paper asks a question that matters beyond Pakistan: when speech and language models are compressed to run on cheap hardware, what do they lose first, and is it the part that can hurt a patient?

Hooker et al. showed that compression hits the rare long tail of the data hardest while headline accuracy barely moves (arXiv). Marchisio et al. found quantization hurts non-Latin-script languages most, and that automatic metrics hide it: a 1.7% automatic drop in Japanese was a 16.0% drop for human raters (arXiv). In a consultation, the long tail is exactly the drug names, the doses and the Punjabi.

1. **RQ1.** How do open ASR models handle code-switched Urdu-Punjabi-English clinical speech, and where do they break?
2. **RQ2.** Does INT8 and INT4 quantization damage drug names, doses and negations faster than overall WER shows, and what is the smallest setup on each device tier that stays safe?
3. **RQ3.** Can a formulary-grounded agent recover those errors without inventing new ones?
4. **RQ4.** In a real OPD, how many edits do doctors make, and how much time does it save?

Contribution	Closest prior work	Gap Nabz fills
Clinical code-switched speech benchmark	Clinera: Egyptian Arabic-English, 683 utterances. UrduSpeech: general Urdu, no clinical category	Urdu-Punjabi-English consultations with paired dictations, tagged for drugs, doses and negation
Clinical risk metric	Clinera's medical-term F1	Adds dose errors and negation flips, weighted by harm
Quantization study	Marchisio et al., text-only LLMs	Speech-to-note pipelines, where two compressed models stack their errors

Contribution	Closest prior work	Gap Nabz fills
Edge deployment frontier	MobileQuant: on-device LLM speed and energy, no speech or clinical task	Accuracy against RAM, latency and battery for full speech-to-note pipelines on phones and old PCs
Formulary-grounded correction	Korean-English dictionary plus LLM post-processing (JMIR 2026)	Agentic, offline, every line linked to audio
Field evidence	MedIQ and Hami: claims, no public benchmark found	Open numbers from real doctors

Data and ethics

The test set stays small and clean: about 10 hours of fully annotated consultations. Clinera built a publishable benchmark from 683 utterances, so scale is not the bar; annotation quality is.

1. **Scenario cards.** With two partner doctors, write about 150 cards from common OPD cases: fever, typhoid, dengue, diarrhoea, hypertension, diabetes, gastritis, chest infection. Cards describe the patient, not the words, so speech stays natural.
2. **Role-play recording.** House officers and final-year students play the doctor. Volunteers play patients and drop into Punjabi where they naturally would. Target 120 to 150 consultations of 4 to 5 minutes, recorded on ordinary phones with OPD-style background noise. After each one, the doctor dictates a 30-second summary, giving paired ambient and dictation audio.
3. **Annotation.** Transcribe in Label Studio and tag speaker turns, language spans, drug names, doses and negations. Never build ground truth by correcting one candidate model's output: UrduSpeech corrected Gemini drafts and then found Gemini best by a wide margin. Draft with a model that is not under evaluation, or transcribe the test split from scratch.
4. **Training data, kept apart from the test set.** Common Voice Urdu, FLEURS, UrduSpeech's 89.4 hours of code-switched speech, and synthetic drug-name sentences voiced by TTS. Synthetic audio never enters the test set.
5. **Formulary.** Build the drug lexicon from DRAP's public registry of brand names, generics, dosage forms and composition (DRAP), and use DrugsInfo for interaction data.

Ethics

- Role-play data involves consenting adults and no patients, so review is light.

- The pilot needs institutional ethics approval for both modes, filed together from day one. Each ambient visit starts with the patient's consent captured on audio, plus written consent during the pilot.
- Government OPD rooms often hold several patients at once, so bystander voices are addressed in the application and the app has a one-tap pause. By default, audio is deleted once the note is signed and only the transcript is kept.
- The study needs a faculty sponsor on the ethics application, and partner doctors to record and judge the system.
- Release audio only from speakers who agreed to public release. Transcripts, tags, models and code can all go public.

Evaluation and pilot

Every model is scored on the same test set at FP16, INT8 and INT4, then timed on all three device tiers.

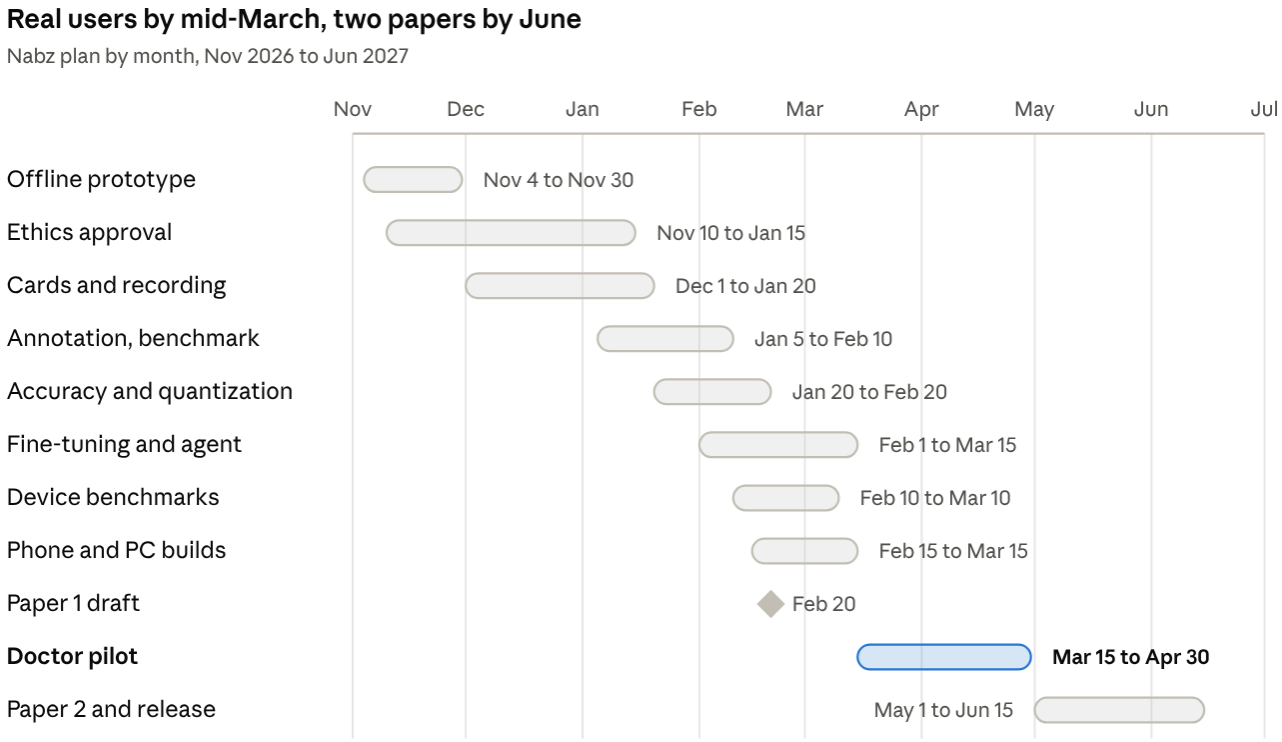
Layer	Metrics	Candidates
Speech to text	WER and CER, split by Urdu, Punjabi, English and switch points	Whisper tiny, base and small, zero-shot and fine-tuned; Whisper large-v3-turbo; Omnilingual ASR 300M and 1B; Omnilingual 7B and Seamless M4T v2 as accuracy ceilings
Critical tokens	Drug-name F1, dose error rate, negation flip rate	Same models at each precision
Note	Modified PDQI-9 rated by doctors, hallucinations per note	Gemma 4 E2B and E4B, Qwen3.5 small models; a large cloud model as the upper bound
Device	Real-time factor, seconds to draft, peak RAM, battery per note	Low-end phone, mid-range phone, old PC
Mode	Accuracy and minutes per note	Dictation against ambient, on the paired recordings
Speaker labels	Doctor or patient attribution error, live transcript lag in seconds	Voice match against full diarization, on the ambient recordings

Pilot

- 5 to 10 doctors for 4 weeks, using both modes, with each doctor choosing per patient. Size is not the bar: Kissan-Dost reported a 90-day pilot with 5 participants (arXiv).

- Run a baseline week first, timing how long notes take by hand.
- Measure edit distance from draft to signed note, minutes per note, how many flags were real errors, and a System Usability Scale survey.
- Report every critical error individually, not just averages. Clinera's best system still made 11 fatal or high-risk errors, affecting 1.6% of its corpus (JMIR Preprints).
- Score notes with the modified PDQI-9 plus hallucination checks, as Palm et al. did on 97 encounters (Frontiers).

Timeline



timeline · Nov 2026 to Jun 2027

The pilot is the gate: without doctors using Nabz by mid-March, Paper 2 has no field evidence. Plan the recording sprint around winter break.

Paper 1, the benchmark and quantization study, targets Interspeech 2027, held 29 Aug to 2 Sep in São Paulo. Its deadline is not announced yet; recent editions fell between 25 February and 21 March (mldeadlines). Paper 2, the full system with device benchmarks and the pilot, goes to an ACL Rolling Review cycle or JMIR Medical Informatics once the pilot data is in.

Risks

The riskiest piece is the slowest device in the pilot, so it gets tested first.

Risk	Fallback
Low-end phones too slow	Live dictation and after-visit ambient on 4 GB phones, smallest models; the draft builds while the next patient sits down
Old PCs lack AVX2 or RAM	Fallback builds; test on the slowest machine at the pilot site first
Phone heats up or drains battery	Short clips and one model in memory at a time; report battery used per note
App too large to download	Ship model files separately by USB or local Wi-Fi
Ethics approval drags	Role-play data needs little; file both modes together, early
Annotation eats the schedule	Cap the test set at 10 hours; two medical students annotate, credited in the paper
Too little Punjabi	Report Punjabi as its own slice; a bad number is still a finding
Doctors drop it after a week	Co-design with two doctors from day one; judge by edit counts, not compliments
An error reaches a patient	The doctor signs every note; the system flags instead of guessing and never suggests a diagnosis
Live transcript lags on mid-range phones	Whisper base drives the live view; small reruns in the background for the final note
Bystander voices in shared OPD rooms	One-tap pause, covered in the ethics application, audio deleted once the note is signed
Patient declines recording	Doctor switches to dictation for that patient

Related work

Nabz sits where four lines of work meet: Urdu speech recognition, clinical speech, model compression, and real deployments in South Asia.

Work	What it shows	Use in Nabz
Antall and Akhtar, 2025	Whisper Small reached 33.68% WER on Urdu with no fine-tuning; Tiny reached 67.08%	Zero-shot baseline for the phone tier
Sehar et al., CHiPSAL 2025	Few-shot fine-tuning clearly improves Whisper on Pashto, Punjabi and Urdu	Fine-tuning recipe with limited data
WER We Stand, COLING 2025	Seamless leads on read Urdu speech, Whisper-large on conversational speech	Model shortlist and Urdu text normalization
UrduSpeech, 2026	Code-switching nearly doubles Whisper-large-v3 WER; 89.4 hours of code-switched audio released	Failure modes and training data
Omnilingual ASR, 2025	1,600+ languages, models from 300M to 7B	Edge-sized recognition candidate
Clinera, 2026	First code-switched medical ASR benchmark (Egyptian Arabic-English), with medical-term metrics	Template for the benchmark and metrics
Korean-English medical ASR, JMIR 2026	Dictionary plus LLM post-processing fixes code-switched medical terms without retraining	Basis for the drug-correction tool
Palm et al., 2025	Modified PDQI-9 with hallucination checks across 97 encounters	Note-quality scoring
Hooker et al., 2019	Compression hurts the rare long tail while headline accuracy holds	Core idea behind RQ2
Marchisio et al., 2024	Quantization hurts non-Latin scripts most, and automatic metrics hide it	Case for doctor-rated evaluation
MobileQuant, 2024	Integer-only quantization cuts on-device latency and energy by 20 to 50%	Phone-tier quantization

Work	What it shows	Use in Nabz
Dhenu2, 2024	A 1B domain model, trained on 1.5M instructions, built for phones	Recipe for a small domain-tuned note writer
FarmerChat, 2024	GenAI advisory deployed in four countries, reaching 15,000+ farmers	How to report a real deployment
Kissan-Dost, 2026	Pakistani WhatsApp voice advisory with a 90-day, 5-participant pilot	Pilot size and reporting in Pakistan
EkaScribe	India's ambient scribe for mixed Hindi-English OPD speech	Closest commercial precedent

Deliverables

1. **Open benchmark:** about 10 hours of code-switched Urdu-Punjabi-English consultations, with paired dictations and drug, dose and negation tags.
2. **Quantized models:** fine-tuned speech recognition and note writer for the phone and PC tiers.
3. **Software:** an offline Android app and a single offline installer for old PCs, both open source.
4. **Device report:** accuracy, latency, peak RAM and battery for every model on three device tiers.
5. **Pilot report:** 5 to 10 doctors over 4 weeks.
6. **Two papers:** the benchmark and quantization study (Interspeech 2027 target), then the full system with device benchmarks and the pilot (ACL Rolling Review or JMIR Medical Informatics).

First milestones

- [ ] Feasibility spike: 10 dictations and 3 role-play conversations through whisper.cpp and llama.cpp at 4 CPU threads and on an Android phone, logging time, RAM and errors on drugs, doses and negations
- [ ] Offline prototype with live transcript in both modes, producing a note on an Android phone by Nov 30, 2026
- [ ] Ethics application filed for role-play recording and both pilot modes
- [ ] Scenario cards drafted with two partner doctors

References

Healthcare context

1. Govt urged to ease OPD burden on doctors. Associated Press of Pakistan, 12 Dec 2024.
2. Surgical OPD consultation study, Civil Hospital Karachi. Pakistan Journal of Medical Sciences, 2009 (2008 data).
3. Raza et al. Prescription patterns of general practitioners in Peshawar, Pakistan. Pak J Med Sci 30(3), 2014.
4. Riaz et al. Medication errors in manual and computerized prescription systems in public hospitals in Lahore. PLoS ONE 9(8), 2014.
5. Haroon. AI in Healthcare: Physician and Patient Survey Report. FAST School of Management, FAST-NUCES Lahore, Oct 2026 (unpublished).
6. DRAP drug database and the duplicate Valtec brand. Dawn, 23 Aug 2015.
7. DRAP registered product data. Drug Regulatory Authority of Pakistan.
8. DrugsInfo. DRAP-aligned drug and interaction reference.

Market

9. MedIQ ambient clinical intelligence. TechJuice, 20 Nov 2025.
10. Hami physician assistant launch. CW Pakistan, 31 Jul 2025.
11. EkaScribe launch. Digital Health News.
12. Mobile OS market share in Pakistan. StatCounter, Oct 2026.

Urdu and clinical speech

13. Nafees ul Haq et al. UrduSpeech: a 156-hour Urdu speech corpus. arXiv, 2026.
14. Antall and Akhtar. Lightweight Whisper models for low-resource Urdu transcription. arXiv, 2025.
15. Sehar et al. Benchmarking Whisper on Pashto, Punjabi and Urdu. CHiPSAL, 2025.
16. Arif et al. WER We Stand: benchmarking Urdu ASR models. COLING, 2025.
17. Omnilingual ASR team. Omnilingual ASR for 1,600+ languages. arXiv, 2025. Model sizes: The Decoder.
18. Behairy and Ahmed. Clinera ASR benchmark. JMIR Preprints, 2026.
19. Postprocessing for bilingual code-switched medical speech recognition. J Med Internet Res, 2026.
20. Palm et al. Assessing the quality of AI-generated clinical notes. Frontiers in AI, 2025.

Compression and edge

21. Hooker et al. What do compressed deep neural networks forget? arXiv, 2019.

22. Marchisio et al. How does quantization affect multilingual LLMs? Findings of EMNLP, 2024.
23. Tan et al. MobileQuant: quantization for on-device language models. Findings of EMNLP, 2024.
24. Google releases Gemma 4. DataNorth, 3 Apr 2026.
25. sherpa-onnx, with supported models listed in the llm_toolkit docs.
26. VOSK supported languages. Alpha Cephei.
27. Local Whisper speech-to-text comparison. PromptQuorum, 2026.
28. Matcha Urdu TTS. Hugging Face.

Deployments

29. KissanAI. Dhenu2 India 1B. Hugging Face, 2024.
30. Singh et al. Farmer.Chat. arXiv, 2024.
31. Ali et al. Kissan-Dost. arXiv, 2026.

Venues

32. Interspeech deadlines. mldeadlines.